Ahraz Kibria

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Summary

Software Engineer building scalable **backend** and **distributed systems**: cut production **data pipeline latency** by **95% (3 hours to 10 minutes)**, integrated telemetry across **500,000+ vehicles (53 databases)**, and built an **event-driven** transit platform. Open to remote and roles across Canada and the US.

Experience

IT Network Engineer **May 2025 – Present**
CAPREIT *Toronto, ON*

- Built **Python** automation scripts and **REST API** integrations on **Linux** for troubleshooting, monitoring, and health-check reporting; evaluated **GitHub Copilot** workflows to standardize team software engineering practices.
- Analyze operational telemetry across Greater Toronto Area properties through the **Cisco Meraki Dashboard** and **REST APIs** to find trends and anomalies and resolve production degradation; enforce data governance, access controls, and network segmentation (**VLANs, firewalls, VPNs**) in **ServiceNow**.

Integrations Engineer **Nov. 2024 – Sept. 2025**
Geotab *Oakville, ON*

- Cut **data pipeline latency** from **3 hours to 10 minutes (95%)** by designing a redundancy-skipping algorithm in **Python** ETL/ELT pipelines on **Google BigQuery (SQL)**, driving backend **scalability** and **performance optimization**.
- Integrated vehicle telemetry from **53 production databases** covering **500,000+ vehicles** with rate limiting, schema validation, and automated **unit and integration testing**, sustaining **99.9% data quality**.
- Built the API Adapter Wizard, an **object-oriented C# / .NET WPF** desktop app (**MVVM**) that replaced hand-edited JSON configuration with a guided tool, achieving **90%+ faster onboarding for 100+ global partners**.
- Diagnosed OEM device activation failures with hypothesis-driven **debugging**, showed they were timing-related rather than logic errors, simplified the backend code, and cut execution time by **80%+**.
- Shipped a production generative AI agent and intelligent search (**Augmented Analyst**: enterprise **retrieval-augmented generation (RAG)** and vector search over terabytes of **BigQuery** data via a **Flask REST API** with guardrails and multi-model fallback) on **Google Cloud (Vertex AI, Gemini, Cloud Run)** used by **2,500+ people**; conducted **code reviews** and mentored a new intern.

Solutions Engineer **Sept. 2024 – Nov. 2024**
Geotab *Oakville, ON*

- Turned business requirements into technical **REST API** specifications, **system design** models, and data deployment architectures with data scientists, engineers, and product teams, and prototyped geospatial analytics visualizations.

Projects

TransiNet – Real-Time Distributed Transit Platform | *Python, FastAPI, RabbitMQ, gRPC, React, Docker* **Mar. 2026**

- Built an **event-driven** system of **4 Python microservices (7 Docker containers on Linux)** that polls the TTC GTFS-Realtime feed every 10 seconds and fans updates out through **RabbitMQ message queues**.
- Engineered a **FastAPI WebSocket** gateway streaming live positions to a 3D **React / deck.gl full-stack** map in **under 100 ms**, with **gRPC / protobuf** bulk history transfer, **MongoDB** for telemetry, **PostgreSQL** for predictions, and **RandomForestRegressor** crowding dispatch recommendations.

Project Aegis – Full-Stack Clinical Platform | **aegishealth.dev** | *Python, Django, React 19, GCP* **Nov. 2025 – Apr. 2026**

- Built a **full-stack** platform combining a **Django REST API** and **React 19** dashboard (5 visual views, OCR/barcode intake, PDF reports) with a **Neo4j** graph (**292,000+ edges**) and **Gemini 2.5 Flash GraphRAG** assistant at **sub-200 ms P95 latency** (**99.6% PR-AUC**, **92% fewer false alerts**).
- Deployed containerized **microservices** in **Docker** via **Google Cloud Build CI/CD** to **Google Cloud Run** and **Kubernetes (GKE)** with automated **unit testing** and **integration testing** (**pytest, Playwright**) and model governance audit logs.

KDCore – Secure Multi-Party Cryptographic Relay | *Java, RSA-2048, AES, SHA256withRSA, Java Sockets* **Mar. 2026**

- Built an **object-oriented Java** Key Distribution Center over raw **Java sockets** authenticating 3 concurrent clients with **RSA-2048** handshakes, nonce challenges, **AES** group session keys, **SHA256withRSA** signatures, and timestamp replay protection, verified via **JUnit unit testing**.

+7 more projects · github.com/ahrazkk

Education

Toronto Metropolitan University (TMU) **Aug. 2021 – May 2026**
B.Eng. Computer Engineering (Software Engineering) · ***Dean's List, Fall 2025 and Winter 2026*** *Toronto, ON*

Skills

Languages: Python, Java, C#, TypeScript, JavaScript, SQL, C, VHDL
Backend: Microservices, Event-Driven Architecture, REST APIs, gRPC, RabbitMQ, WebSockets, Django, FastAPI, Flask, .NET
Cloud / DevOps: Google Cloud (Cloud Run, GKE, Cloud Build), AWS, Docker, Kubernetes, Linux, CI/CD, Pytest, JUnit, Playwright
Data & Frontend: PostgreSQL, MongoDB, Google BigQuery, Neo4j, ETL/ELT Pipelines, React 19, C# .NET WPF (MVVM), deck.gl, WebGL